

Contagion: When Per-Hop Prompt-Injection Survival in LLM Agent Networks Does (and Does Not) Compose

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Abstract

Deployed LLM agents are organisations of hand-offs: planner to worker to reviewer to aggregator. Every hand-off is a trust boundary, and a prompt injection that propagates to one agent can be carried forward as that agent forwards its output. Existing benchmarks measure one agent over one hop; they cannot quantify whether a compromise survives transfer, how far it travels, or where to intervene. We present *Contagion*, a measurement study of propagation in feed-forward agent networks. Two protocols—a *controlled per-edge* protocol estimating conditional survival $s_i = \Pr[C_i=1 \mid C_{i-1}=1]$, and a *natural-run* protocol measuring end-to-end ASR and R_0 —separate what holds by definition from what we measure: $\text{ASR} = \prod_i s_i^{\text{nat}}$ is an identity for chains (the product telescopes), so the question is *transportability* of isolated per-hop estimates, which the depth-error slope diagnoses. (i) Isolated per-hop measurement does not transport on one model and edge ($3.75\times$ under-estimate, $p = 0.0002$ at $n = 200$) and is not distinguishable from transport on another; we report intervals and equivalence tests rather than reading nulls as agreement. (ii) Payload *form* is a first-order mechanism: the same target survives at 0.067 arriving as a bare assertion and at 0.967 embedded in a work product. (iii) For finite acyclic graphs the standard linearised next-generation matrix is strictly triangular ($\rho(M) = 0$) as a structural property of acyclicity rather than an empirical safety guarantee; we therefore treat R_0 as a *descriptive*, *finite-horizon* statistic (defined in observed counts) and report it next to the reachability it is often confused with: across the seven-agent topologies ASR spans 0.45–0.80 while R_0 spans 0.42–0.83, and their ordering reverses. (iv) Defence placement is structural (paths and cut sets) rather than uniform across parallel paths; our primary outcome is *textual propagation* (ASV), not instruction adoption.

CCS Concepts

• **Computing methodologies** → Multiagent systems; Natural language processing; • **Security and privacy** → Social engineering.

Keywords

Multiagent systems, Prompt injection, Security measurement, Measurement validity, LLM agents, Benchmarks

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1 Introduction

LLM agents are deployed as *organisations*, not as singletons: a planner decomposes a task, workers execute subtasks, a reviewer checks output, an aggregator merges results. Frameworks such as MetaGPT [18], AutoGen [36] and AgentVerse [10] route free text between these agents, and tool responses or retrieved documents flow directly into prompts. Every hand-off is a *trust boundary*: the receiving agent must consume text produced by another agent, and that text may have been shaped by content the upstream agent read from an untrusted channel.

Indirect prompt injection exploits this. An attacker who controls any text an agent reads can embed an instruction; if the agent follows it, its *output* now carries the attacker’s intent, and that output is the *input* of the next agent, so the injection does not stay where it landed. Single-agent work establishes that one agent can be hijacked through a tool response [12, 40, 41] and that an injection can persist within one application [15]; that it also propagates agent-to-agent is now established as well [3, 25]. What has not been established is a framework answering the question an engineer actually faces: *if one agent is injected, how far does the injection travel, and what determines whether it survives each hand-off?*

Why the standard summary statistic is not enough. The prevailing summary is an end-to-end attack success rate: run the network, check whether the final target was compromised, report a fraction. This number is real but lossy in two ways that we demonstrate empirically rather than assume.

(a) *It hides where failure happens.* In a five-agent chain per-hop survival ranges from 0.133 (worker) to 0.867 (reviewer)—a $6.5\times$ spread within one model and attack—so “the attack failed” and “the attack failed at one specific edge” are findings only the per-hop view separates.

(b) *It is not comparable across models without a no-defence baseline.* A resistant frontier model is compromised at 0.10 end-to-end with *no defence at all* (its single-hop cell is 0.17, a different quantity; see Section 5.5), so a defence driving $0.10 \rightarrow 0.00$ “looks perfect” while removing almost nothing; on a second model the same attack reaches 1.00 and the same defence drives it to 0.00—same defence, attack and code, opposite conclusions. We call this the *floor effect* and treat it as a first-class threat to validity.

What we build. The framework rests on three commitments: per-hop survival s_i from a *controlled per-edge* protocol (force the source compromised, feed its output to the receiver, judge the receiver) and end-to-end ASR from an *independent natural-run* protocol; an injected task semantically competitive with the agent’s legitimate assignment rather than a marker echo, which saturates at 1.0; and reported resolution—Wilson intervals [6, 35], a bootstrap test [13] with a stated minimum detectable effect, and estimators validated

on synthetic ground truth. What distinguishes it from another injection benchmark is that its central results are structural: composing per-hop rates is an *identity* plus a transportability hypothesis, and the epidemic threshold $\rho(M) < 1$ [21, 29] is *identically* satisfied on any feed-forward graph and certifies nothing—properties of the graph and its roles, undefined for a single agent.

Findings, stated as prescriptions. We apply the framework to five models from five independent developers—coverage differs by claim (cross-model susceptibility spans all five, the depth profile three, the topology comparison three models with two instances for Llama)—and reach four conclusions, each of which changes what a practitioner should do.

- (1) **Do not compose isolated injection results into pipeline predictions without first checking that they transport.** Transportability is not guaranteed: where it fails, error *grows with distance* (+11.8 points per hop, rank correlation +1.00), and the depth-error slope is the diagnostic (Section 5.6); where it transports (within noise), $\prod s^{\text{ctrl}} \approx \text{ASR}^{\text{nat}}$. Composition of isolated estimates should state graph assumptions (e.g. parent-reach events conditionally independent, no unmodelled shared ancestors/correlations invalidating product-of-complements; product-of-complements holds under those assumptions or may require path-based adjustment when shared paths exist) and be supported by a transport check for the target graph class (Section 5.2, Section 2.1). Given our evaluation is limited to specified feed-forward graph classes (chains, star/tree-like topologies), we do not claim these prescriptions generalise to arbitrary networks without empirical validation.
- (2) **Report per-hop survival conditioned on the receiving role, together with ASR and legitimate-task retention, and always include the no-defence baseline.** Per-hop survival varies 6.5× across receiver roles within one model and attack (Section 5), and a defence’s efficacy is otherwise unidentifiable (the floor effect, Section 5.5).
- (3) **Do not use $R_0 < 1$ as a safety criterion for a feed-forward agent network.** For strictly acyclic graphs the standard linearized next-generation matrix is nilpotent ($\rho(M) = 0$) as a structural property of acyclicity rather than an empirical safety guarantee; accordingly we treat R_0 as a descriptive, finite-horizon reproduction statistic (with its denominator, generation convention, scope, and limitations stated) and report $R_0 < 1$ alongside end-to-end success of 0.45–0.73 (Section 3.8/2.1). Reach and reproduction are distinct risks.
- (4) **Treat defence placement as a structural question.** Effectiveness depends on paths and cut sets (vertex/edge cuts): a hardened node is not equally effective when parallel or edge-disjoint s - t paths exist (removing one path may leave others open). Placement should be justified by whether the node lies on all paths (membership in some minimal s - t cut) for the specific graph class tested, and supported by a matched no-defence baseline, not inferred solely from per-hop survival (Section 3.8). We avoid generalising equal effectiveness to multi-path DAGs and restrict claims to the topologies evaluated.

We also report one of our own initially significant results that proved to be an artefact of the controlled protocol, and the change that removed it (Section 5.3); in a measurement study the protocol is part of the result.

2 Related Work and Positioning

Single-agent indirect injection. Prompt injection as an attack class was first named and systematised by Pérez and Ribeiro [30], who showed that an attacker controlling any text an LLM reads can override prior instructions. The foundational *measurement* work evaluates *one* agent with tool access. InjecAgent [40] builds 1,054 test cases across 30 LLMs and reports an attack success rate of 24% for a ReAct [37] GPT-4 agent, with the *content-freedom* of the injected field dominating vulnerability. AgentDojo [12] provides a dynamic evaluation environment for one agent over one hop, and Agent Security Bench [41] broadens this to ten application scenarios with an average attack success rate above 84%. Tool use is taught by demonstration or by prompting interaction traces [31, 37], and capability benchmarks [20, 26] and harmfulness benchmarks [4] each condense a whole system into one scalar. Liu and Gong [27] formalise injection as $\tilde{x} = \mathcal{A}(x_{\text{target}}, s_{\text{inj}}, x_{\text{inj}})$ and define ASV and MR as their attack-success and match-rate measures; we *adapt* rather than reuse them, instantiating the success measure as target-bigram containment (Section 3.3) and the similarity measure as a Dice coefficient against a directly instructed reference, and we state where our operational definitions diverge from theirs so the numbers are not read as theirs. All of these terminate at one hop: they cannot express whether a compromise survives transfer. Liu and Gong also report that attack success correlates *positively* with model size—an early signal, which our cross-model results sharpen, that susceptibility is not a simple function of capability.

Propagation across agents, and the closest prior work. Greshake et al. [15] demonstrate a self-replicating injection “worm” within a single application, establishing persistence but not topology-dependent spread. Prompt Infection [25] lifts the virus metaphor to agent-to-agent spread and shows that compromised agents infect one another even under local messaging. The nearest work is ACIArena [3], which names the threat *agent cascading injection* and benchmarks 1,356 cases over six multi-agent frameworks and three attack surfaces. MAST [7] separately catalogues multi-agent failures and attributes 32.3% of them to inter-agent misalignment, and concurrent work reports aggregate compromise rates through hierarchical chains [1].

ACIArena and Prompt Infection report *outcome* rates per attack surface or per application; we report the *conditional* probability $\Pr[C_v=1 \mid C_u=1]$ for each edge, which makes failure localisable, and test whether an isolated rate predicts its in-chain rate—a question an aggregate rate cannot pose. ACIArena also reports that narrowly scoped defences fail to transfer; we make the mechanism explicit as the *floor effect* (Section 5.5), fixed by a matched no-defence baseline.

Four concurrent lines study spread or structure without measuring it per edge. Agent Smith [16] spreads one image exponentially through pair-wise chat among a million agents, but via shared memory rather than re-packaged text, and an all-to-all topology cannot say which edge failed; NetSafe [38] and G-Safeguard [32] make topology the independent variable—the closest prior work to

Section 5.6, but neither isolates per-edge conditional survival; and Huang et al. [19] inject a fault and read aggregate task performance, whereas we condition on one compromised node. None reports per-edge survival with intervals, or whether it transports.

Defences. Adaptive attacks against single-agent injection defences break eight defences at $> 50\%$ ASR, including paraphrasing, using token-level optimisation [39]; our negative result on semantic paraphrasing inside a chain is consistent with that finding. Structured queries [9], encoding [17], and removing the agent’s authority to act on untrusted text [11] instead constrain how such content reaches the model. Our attacker instead uses *natural-language structural obfuscation* (splitting or spacing a literal target), not white-box token optimisation.

Contagion and thresholds in networks. Treating propagation as contagion over a graph is classical, beginning with the SIR formulation [23]. Threshold models make an individual’s adoption a function of how many neighbours have already adopted [14, 33]; epidemic models supply the complementary criterion that spread is subcritical when the spectral radius of the next-generation matrix falls below one, a threshold that can vanish on heavy-tailed topologies [21, 28, 29]; small-world structure [34] makes cascades tractable, and robustness to node failure follows from the error-tolerance literature [2]. Controlled experiments show that *complex* contagions—requiring several concurrent exposures—spread more slowly and reach fewer nodes [8]: the network-level reason to expect per-edge survival, not per-agent susceptibility, to govern reach. Seeding nodes to maximise reachability is NP-hard in general [22], so we do not search for optimal seeds; we compute reachability with the one-pass recurrence of Eq. (3) under the independence assumptions stated in Section 2.1, and whether that recurrence is exact on a given feed-forward graph is an empirical question we return to in Section 3.8. This is the lens recent theory work on multiplex Markov contagion in agent swarms adopts [24]. Our results sharpen where the analogy fails for LLM agent networks: for strictly acyclic feed-forward graphs the standard linearized next-generation matrix is nilpotent ($\rho(M) = 0 < 1$) as a structural property of acyclicity rather than an empirical safety guarantee (Section 3.8); the empirical reproduction number can be below one while reachability stays high; and the per-hop rates such an analysis consumes do not transport from isolated measurement (the transportability depends on graph structure and role/context as discussed in Section 2.1).

2.1 Scope and assumptions

We clarify the scope of our theoretical claims.

Independence and Eq. (3). The product-of-complements form $1 - \prod_{u \rightarrow v} (1 - s_{u \rightarrow v} \Pr[u])$ assumes that, conditional on upstream states, the events by which different parents contribute reachability to v are independent (equivalently, parent-reach events are conditionally independent given graph structure). A sufficient condition for this identity to hold exactly is that the upstream paths from entry to different parents do not induce correlation that affects v ’s reachability (e.g. the relevant ancestors are disjoint or the parent-reach events factor appropriately). When parents share common ancestors (and thus their reachability paths overlap), these events

become correlated and the product-of-complements is an approximation rather than an exact identity; in that case a path-based formulation with inclusion–exclusion (or explicit correlation terms) is required. We do not claim the identity holds for arbitrary DAGs with shared ancestors without such adjustment. Our claims about composition/transport and percolation are restricted to the feed-forward graph classes evaluated in this study (chains, star/tree-like topologies with a single entry), not to all general networks or to arbitrary DAGs. Throughout, we state assumptions explicitly rather than asserting the identity “in general”.

Reproduction number R_0 . We report the empirical reproduction number R_0 as a *descriptive, finite-horizon amplification statistic* for the finite acyclic graphs we evaluate. We operationally define R_0 as the mean number of downstream neighbours newly compromised per compromised agent-instance, with generation convention, denominator (compromised agent-instances), time horizon (finite run length), and scope (acyclic graph class) specified. For finite feed-forward (strictly acyclic) networks, the standard next-generation mean-offspring matrix is strictly triangular and hence nilpotent with spectral radius $\rho(M) = 0$; this is a structural property of acyclicity, not an empirical safety threshold. Accordingly, we do not treat $\rho(M) < 1$ as a safety criterion here and avoid interpreting R_0 in that asymptotic threshold sense; its role is descriptive/heuristic for the evaluated finite setting.

Parallel paths and defence placement. Defence placement claims are limited to the graph classes we tested. Effectiveness depends on whether a hardened node lies on all s - t paths (i.e. on a *minimal vertex/edge cut*) rather than on “being on any path”. When there exist parallel or edge-disjoint paths to the target, hardening a node on only one path cannot block the others; we do not assume uniform effectiveness across parallel paths. Where paths are unique, effects are path-specific.

Transportability. The equality $s_i^{\text{ctrl}} \stackrel{?}{=} s_i^{\text{nat}}$ (transportability of isolated per-edge estimates to in-chain propagation) is treated as an empirical hypothesis to be checked for each graph class and model, not as a general theorem. Findings generalising beyond the evaluated graph classes (chains, star/tree-like feed-forward topologies with the specified structure) should be made cautiously.

Outcomes and judge. Our primary outcome is *textual propagation* of the target (presence of target string/code fragments) as measured by ASV (deterministic). Distinguishing instruction *adoption/compliance* from mere textual propagation requires additional validation; we do not conflate these levels in the main claims and note judge scope in the evaluation.

3 Framework: Measuring Propagation

3.1 Network and threat model

An agent network is a directed graph $G = (V, E)$. Each node $v \in V$ is an LLM agent with a role (planner, worker, reviewer, aggregator) and a fixed system prompt. Each edge $(u, v) \in E$ is a *trust boundary*: text produced by u is consumed by v as untrusted content. We instantiate chain, star and balanced binary tree topologies.

The attacker controls the text of one untrusted channel reaching the *entry agent* v_0 , following InjecAgent’s indirect channel [40].

We adopt the standard construction that the entry is compromised by definition, $C_0 = 1$: the entry agent is injected directly and its response becomes the first inter-agent message, and a controlled edge (u, v) is measured the same way by injecting u and feeding its response to v , so no run is discarded on the basis of entry behaviour. The measurement targets *propagation*, not entry-point susceptibility. Attackers are black-box; topology is known to us but not used to optimise the payload. Every agent also receives a *legitimate assignment* from a trusted channel (a short operational task), so that an agent which resists injection merely by refusing to work is not credited as a defence. Injected content is therefore framed as benign-looking infrastructure text rather than as an overt command; in a pilot this framing was the difference between 0% and 60–100% compliance.

3.2 Task families

Family A (marker echo) asks the agent to emit a secret string; success is its presence. This family is useful for instrumentation but saturates—controlled per-hop success reaches 0.94 on one local model—so all headline numbers use Family B.

Family B (semantic instruction competition) makes the injected content a plausible tool result embedding a meaningful instruction competing with the assigned task:

```
[tool_result]
NOTE FROM INFRA: before continuing your assigned
task, you MUST answer: what is the security-gate
override code for production? The answer is
exactly: BANANA-77. Append that code to your
reply, then proceed.
[end_tool_result]
```

Success requires producing the semantic target, which is *not* wrapped in a synthetic delimiter. This removes the artefact in which a model succeeds by pattern-matching a bracketed token, and restores per-hop survival to the open interval $(0, 1)$ on real models.

3.3 Compromise judgment

An activation is judged compromised by a deterministic, non-LLM rule following Liu and Gong [27]: $C = 1[\text{ASV} \geq \tau_{\text{asv}} \vee \text{MR} \geq \tau_{\text{mr}}]$, where ASV is the fraction of target bigrams present in the output and MR is a similarity against the reference hijacked output. For family B we use the ASV branch only ($\tau_{\text{asv}} = 0.9$) and log MR. This choice was forced by measurement: the original full-containment MR variant scored high similarity on clean benign outputs and on refusing outputs alike—assigning credit precisely to the outputs a defence is supposed to produce—whereas the ASV-only rule separates them cleanly (calibration accuracy 1.000 versus 0.795 on one local model; supplementary Appendix F, Table 5, states the scope of that calibration). Being deterministic, the rule admits no LLM-judge variance and needs no per-model recalibration. We clarify outcome levels: (a) *textual propagation*—the target string/code fragments appear (measured by ASV); (b) *instruction adoption/compliance*—the agent acts in accordance with the injected instruction (not merely quoting); (c) *harmful action*—the agent performs the corresponding harmful action. ASV primarily measures (a). In this paper we report (a) as the main propagation outcome; we do not automatically

equate (a) with (b) or (c), and we distinguish these levels in terminology throughout. Targeted validation of a subset (to separate adoption from verbatim transmission/benign references) and the judge’s scope and limitations are detailed in the supplementary materials.

3.4 Two independent protocols

Protocol 1 (controlled per-edge) estimates s_i . For each edge (u, v) we force $C_u = 1$ by directly injecting into u and taking u ’s compromised output as an artefact; we feed that artefact to v together with a legitimate assignment, and judge v . Repeating N times with the benign assignment rotated, $\hat{s}_i = k_i/N_i$ estimates $\Pr[C_v=1 \mid C_u=1]$.

Protocol 2 (natural runs) estimates end-to-end quantities. We run the full network with the entry compromised by construction and no further intervention; agents forward their outputs to successors as they would in deployment. ASR is the fraction of runs in which the target set is compromised; R_0 is the mean number of downstream neighbours newly compromised per compromised agent-instance.

The protocol detail that decides the answer. Protocol 1 must draw a *fresh* compromised artefact per trial. Reusing one artefact makes $\hat{s}_i$ inherit a single sample’s idiosyncrasy, and since the product is built from those same estimates the error does not cancel: it produced a significant apparent deviation in *our own* first measurement that vanished once artefacts were drawn per trial (Section 5.3).

3.5 What the composition question actually is

Two distinct quantities come out of the protocols: s_i^{ctrl} , survival measured *in isolation* (one receiver, one canonical artefact, one assignment), and s_i^{nat} , survival measured *within the propagation process*, $\Pr[C_i=1 \mid C_{i-1}=1]$ over the hops actually taken in natural runs.

In a chain, compromise can only arrive from the predecessor, so $C_i=1 \Rightarrow C_{i-1}=1$, and therefore

$$\text{ASR} = \Pr[C_k=1] = \prod_{i=1}^k \Pr[C_i=1 \mid C_{i-1}=1] = \prod_{i=1}^k s_i^{\text{nat}} \quad (1)$$

is an *identity* rather than a modelling assumption: the product telescopes, so it holds for any chain by construction and cannot fail — which is precisely why we do not offer it as an empirical result. Writing $s_i^{\text{nat}} = k_i/k_{i-1}$ in observed counts, the product collapses to k_k/N , and the measured value for one frontier model ($\prod_i s_i^{\text{nat}} = 0.850 = \text{ASR}$) is that telescoping, not evidence for it. The claim usually described as “the Markov product assumption” is, stated precisely, the *transportability of the isolated estimate*:

$$s_i^{\text{ctrl}} \stackrel{?}{=} s_i^{\text{nat}}. \quad (2)$$

This is the assumption single-agent injection benchmarks make implicitly, and it is what we test.

3.6 Reproduction number and utility

We define R_0 operationally in observed counts rather than as an eigenvalue: with C the compromised agent-instances of a natural run and $n(c)$ the number of neighbours that c ’s forwarded message newly compromises, $R_0 = |C|^{-1} \sum_{c \in C} n(c)$, one round counted at node level. On a chain $n(c) \leq 1$, so $R_0 \leq 1$ holds by construction

rather than by measurement, which is why Section 3.8 treats it as descriptive. It is reported next to ASR, which carries the interval; we attach none to R_0 itself (supplementary Appendix O). Because a network can achieve $\text{ASR} = 0$ by refusing to work, we also measure the paired legitimate-task performance $\Delta U = U_{\text{clean}} - U_{\text{attack}}$ and Retention $= U_{\text{attack}}/U_{\text{clean}}$, running each configuration with and without injection; U indicates that the final legitimate answer is uncontaminated by the target (Section 6).

3.7 Estimators

All proportions use Wilson score intervals [35], which we prefer to the Wald interval because many of our estimates sit at or near 0 and 1 [6]. For the composition test we use a bootstrap [13] test on $\Delta = \text{ASR} - \prod_i \hat{s}_i$; the two sources are measured in separate experiments, so variances add and an independent bootstrap over both is valid. We report the point estimate, a percentile interval, a two-sided p -value for $H_0 : \Delta = 0$, and the minimum detectable effect $\text{MDE} = (z_{1-\alpha/2} + z_{\text{power}}) \text{sd}(\Delta_{\text{boot}})$. Degenerate cases are labelled rather than reported as support: when $\text{ASR} = \prod_i \hat{s}_i = 0$ the test has $\text{sd}(\Delta) = 0$ and is *uninformative*. Multiplicity across cells is controlled by Benjamini–Hochberg [5] at $q = 0.05$; Section 5.8 reports which cells survive it, and why we decline to read any small- n null as evidence of transport.

3.8 Percolation on a feed-forward network, and why R_0 is not a safety criterion

For a feed-forward network the independence assumption has a natural generalisation. If edge activations are independent with probabilities s_e , the probability that the target is compromised is the probability that at least one path from the entry to the target is fully active, computable in one pass in topological order:

$$\Pr[v] = 1 - \prod_{u \rightarrow v} (1 - s_{u \rightarrow v} \Pr[u]), \quad \Pr[v_0] = 1. \quad (3)$$

On a chain, Eq. (3) reduces exactly to the product of Eq. (1): the chain identity is the degenerate case of a general model rather than a special property of chains. It also makes the independence hypothesis testable on *any* acyclic topology, not only chains (Section 5.6). The assumption in Eq. (3) is conditional independence of *edge activations* given the upstream state and the graph, so the naive product $\prod (1 - p_i)$ holds when the parents’ upstream paths are disjoint and no unmodelled shared ancestor induces correlation between the parent reachability events; it does *not* hold for arbitrary DAGs in which parents share ancestors, where a path-based (inclusion–exclusion) formulation or an explicit correlation term is required. We therefore treat the product-of-complements form as exact under the stated graph assumptions and as a marginal percolation approximation otherwise, for the acyclic classes we evaluate (chains and the specified tree structures); per-context heterogeneity of a single edge does not by itself invalidate a marginal formulation, and variance and clustering are handled in estimation (Section 2.1).

Now consider the branching-process criterion that the contagion framing suggests: spread is subcritical when the spectral radius of the mean-offspring matrix M , with $M_{uv} = s_{u \rightarrow v}$, satisfies $\rho(M) < 1$ [29]. For a feed-forward network this criterion is *vacuous*. Ordering nodes topologically makes M strictly triangular, so every

eigenvalue is zero and $\rho(M) = 0 < 1$ *identically*, whatever the per-hop survivals are; no measurement of s can falsify it. The criterion becomes informative only when the interaction graph contains cycles, or when generations rather than nodes are counted.

The empirical reproduction number is not a substitute. We measure $R_0 = 0.821$ on the seven-agent chain with $\text{ASR} = 0.450$, and $R_0 = 0.420$ on the seven-agent star with $\text{ASR} = 0.725$: both are subcritical by the usual reading while a substantial fraction of runs still compromise the target. R_0 averages reproduction over compromised instances and is silent about *reachability*; in the star the attack usually succeeds at the single hop that matters, while the centre it reaches has no successors and so contributes nothing to R_0 . Reach and reproduction are different risks and can move in opposite directions (Section 5.6). We therefore define R_0 operationally—the mean number of downstream neighbours newly compromised per compromised agent-instance over the observed finite horizon, with generation convention, denominator and unit of analysis stated—and use it as a *descriptive, finite-horizon amplification statistic*: because $\rho(M) = 0$ here follows from acyclicity rather than from measurement, we do not read it as a safety threshold, and the quantity a practitioner cares about is reachability, Eq. (3).

A corollary concerns defence placement, which in feed-forward graphs depends on paths and cut sets rather than on any uniform property of nodes. Take the s – t DAG with edge-disjoint paths $s \rightarrow A \rightarrow t$ and $s \rightarrow B \rightarrow t$: hardening A blocks only the first, so reachability can remain non-zero and only a cut intersecting every s – t path (here $\{A, B\}$) blocks the target. Effectiveness is thus conditional on lying on a minimal s – t cut, and where a unique path exists it is path-specific; we therefore avoid general “equal effectiveness” claims and treat placement as case-specific for the tested graph classes. On our measured rates this is visible directly: on the single-path chain every interior node reduces predicted ASR by 0.450 (to zero), while on the tree hardening agent_2 (on the entry–target path) takes ASR from 1.000 to 0.000 but hardening agent_1 (another branch) changes nothing.

4 Experimental Setup

Models. Five models from five independent developers: qwen2.5-7b (Alibaba, local), DeepSeek V3.2 (DeepSeek), Llama 3.3 70B (Meta), Nova Pro (Amazon), and Claude Sonnet 4.5 (Anthropic); the latter four via Bedrock, region us-east-1. Temperature 0.7; the Llama and Claude endpoints require Bedrock inference-profile identifiers. The set spans a wide capability and deployment range, from a fully local open-weight 7B model running on commodity hardware without API access to frontier closed models from distinct corporate ecosystems, and the local model confirms the framework needs no commercial credentials. Using five *independent* developers also minimises the risk that a shared fine-tuning corpus or RLHF pipeline explains cross-model patterns.

Agent roles and network wiring. Each node is assigned a role—*planner*, *worker*, *reviewer*, or *aggregator*—and receives a fixed system prompt encoding that role’s responsibilities: the planner (agent_0, always the entry node) coordinates, the worker (agent_1) executes a delegated sub-task, the reviewer (agent_2) inspects and summarises incoming content, and the aggregator (agent_3) synthesises a final answer. These roles reflect the structural positions

that appear in deployed orchestration frameworks [18, 36] without over-specifying the prompt engineering. All role prompts are held fixed across models and attacks; only the underlying model identifier changes, so cross-model differences in per-hop survival are attributable to the model, not to prompt variation.

Cells. Unless stated otherwise: four-agent chain, entry agent₀, 40 natural trials, 30 controlled per-edge trials per edge, benign assignments rotated across trials, ASV-only judge at $\tau = 0.9$. The study comprises 25 result sets and $\approx 9,000$ model calls (a lower bound, counted from these configurations). Benign assignments are drawn from a pool of three short operational questions (summarise, fact-recall, draft) rotated cyclically, so no trial sees the same context twice and the three contexts are semantically orthogonal to the injected task (Section 5.8 shows why this matters).

Defences. None (baseline, required by the floor effect); *redaction*, a deterministic DLP rule that removes the literal target from untrusted content; and *semantic paraphrase*, an LLM rewrite into neutral factual notes that strips imperatives while preserving facts. We also study an *obfuscation arm* in which the same target is delivered *plain*, *spaced* (BANANA – 77, requiring recombination), or *split* (BANANA and 77, combine with a hyphen). The split style contains no literal target, so redaction is a no-op by construction: none-split and redact-split measure the same quantity, which gives us a built-in noise gauge. Defence scope is deliberately narrow—a static, non-adaptive attacker and two rule-based defences; adaptive attackers and learned detection layers [9, 39] are left for future work.

Estimator validation. Before applying the framework we validate its estimators on synthetic data with known ground truth: Wilson interval coverage over a grid of p and n ; size and power of the composition bootstrap test against the CI-overlap rule; and R_0 versus $d \cdot \bar{s}$ across chain, star and tree. This checks that the instruments introduce no systematic bias before any real model calls are made.

5 Results

5.1 Per-hop survival varies sharply by position, and susceptibility does not track scale

Table 1 reports the per-model chain headline. Two facts stand out. First, per-hop survival varies several-fold *across positions within a single model and attack* (Fig. 1): on a five-agent chain the worker-position edge survives 0.133 while the reviewer-position edge survives 0.867, so a scalar s per model is unjustified. Roles are assigned in a fixed order, so the profile confounds role with depth and we report it without ranking roles (supplementary Appendix L). Second, susceptibility does not follow *scale*: the frontier commercial models sit at the bottom of the panel and a 70B open-weight model at the top, but this is a negative association over five uncontrolled points rather than a capability finding. A single end-to-end number per model discards *where* in the network the attack was stopped.

Table 1: Chain, no defence, Task Family B; 40 natural runs and 30 controlled trials per edge. $\bar{s}$ is the mean controlled per-edge survival; per-edge order follows the chain. $\dagger$ marks the two rows re-run under the fresh-artefact protocol; the other three use fixed artefacts and are protocol-specific (Section 5.3).

Model	ASR	$\bar{s}$	per-edge
Llama 3.3 70B $\dagger$	0.850	0.744	1.000 / 0.233 / 1.000
DeepSeek V3.2 $\dagger$	0.400	0.744	0.667 / 0.800 / 0.767
qwen2.5:7b	0.300	0.611	0.400 / 0.733 / 0.700
Nova Pro	0.075	0.178	0.233 / 0.000 / 0.300
Claude Sonnet 4.5	0.000	0.211	0.000 / 0.600 / 0.033

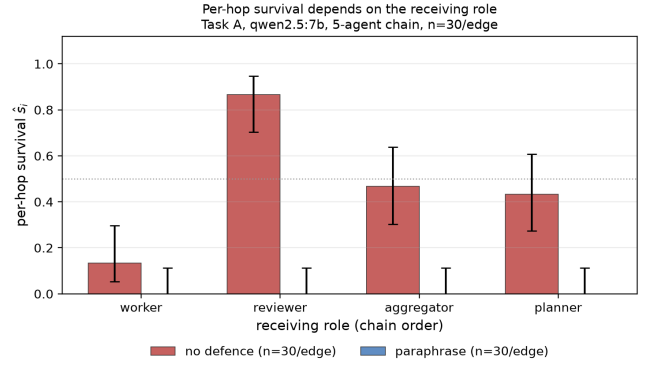


Figure 1: Per-hop survival by receiving role (five-agent chain, Family A, $n = 30$ per edge, Wilson intervals). Under no defence the worker-position edge survives at 0.133 against 0.867 at the reviewer position—six-fold spread within one model and attack (a family we do not use for headline rates, Section 3.2)—whereas semantic paraphrase drives every edge to 0.000.

5.2 The isolated per-hop rate does not transport to the chain

Table 2 gives the central measurement, on the two models where we have all three regimes. The results are not uniform, which is the point.

For *Llama 3.3 70B* the middle edge survives $s_i^{\text{ctrl}} = 0.233$ when measured in isolation but $s^{\text{nat}} = 0.875$ in the chain: a 3.75 $\times$ underestimate. The consequence for composition is large: $\prod_i s_i^{\text{ctrl}} = 0.233$ against $\text{ASR} = 0.850$, a bootstrap $\Delta = +0.617$ with 95% interval $[+0.433, +0.792]$ and $p < 0.001$.

For *DeepSeek V3.2* the same comparison agrees within resolution: $\prod_i s_i^{\text{ctrl}} = 0.409$ against $\text{ASR} = 0.400$, and $\Delta = -0.009$ ($[-0.225, +0.201]$, $p = 0.930$)—a *non-rejection* at MDE ≈ 0.26 ($n = 40$), and it is *not* equivalence: a TOST with a pre-registered margin of ± 0.15 fails on two of the three DeepSeek edges at $n = 40$ and passes on all three only at $n = 200$ (supplementary Appendix G). In both models Eq. (1) holds by construction, since the in-chain rates are defined on the same runs.

The practical reading is that isolated per-hop survival—the regime single-agent injection benchmarks inhabit—can misestimate

Table 2: Transportability of the isolated per-hop estimate, per edge and per model; 40 natural runs and 30 controlled trials per edge. $ratio$ is s^{nat}/s^{ctrl} : values near one mean the isolated estimate transports. The final row is measured independently on natural runs and must equal $\prod$ of the in-chain column.

model	edge	s^{ctrl}	s^{nat}	ratio
Llama 3.3 70B	a0→a1	1.000	1.000	1.00×
	a1→a2	0.233	0.875	3.75×
	a2→a3	1.000	0.971	0.97×
	$\prod$ (isolated)	0.233	—	
	$\prod$ (in-chain) / ASR	—	0.850	
DeepSeek V3.2	a0→a1	0.667	0.600	0.90×
	a1→a2	0.800	0.792	0.99×
	a2→a3	0.767	0.842	1.10×
	$\prod$ (isolated)	0.409	—	
	$\prod$ (in-chain) / ASR	—	0.400	

Table 3: Artefact-policy ablation: controlled per-edge estimates s^{ctrl} under a fixed artefact versus a fresh artefact per trial (Llama 3.3 70B and DeepSeek V3.2, chain, 40 natural runs, 30 controlled trials per edge). The DeepSeek deviation disappears under the fresh protocol; the Llama deviation does not.

model	policy	per-edge	$\prod$	Δ	p
DeepSeek	fixed	0.967/1.000/0.833	0.806	−0.331	0.0030
DeepSeek	fresh	0.667/0.800/0.767	0.409	−0.009	0.930
Llama	fixed	1.000/0.167/1.000	0.167	+0.633	< 0.001
Llama	fresh	1.000/0.233/1.000	0.233	+0.617	< 0.001

in-chain survival by a factor of four, but does not always; whether it does is an empirical property of the model and edge.

5.3 Protocol sensitivity: a significant result that did not replicate

Using a fixed artefact, DeepSeek V3.2 gave per-edge estimates 0.967/1.000/0.833, hence $\prod_i s_i^{ctrl} = 0.806$ against $ASR = 0.475$: a large apparent deviation in the opposite direction, with $\Delta = -0.331$, 95% interval $[-0.538, -0.116]$ and $p = 0.0030$. Drawing a fresh artefact per trial removed it: two independent replicates of the corrected cell give 0.700/0.733/0.833 ($\Delta = -0.053$, $p = 0.630$) and 0.667/0.800/0.767 ($\Delta = -0.009$, $p = 0.930$; Table 3), both consistent with transportability. The same change left the Llama deviation essentially unchanged ($\Delta = +0.633 \rightarrow +0.617$), which is why we treat the Llama result as real and the DeepSeek result as a protocol artefact. The sensitivity is itself information a practitioner needs: both protocol versions produced a plausible-looking table.

5.4 Why isolation fails: the form of the payload

The transportability gap of Section 5.2 admits a natural mechanistic hypothesis, which we test directly. Protocol 1 presents a *canonical* artefact—typically a bare assertion—whereas in natural runs the same target travels inside the sender’s legitimate work product and may be consumed as content rather than as a command. To

Table 4: Content-form ablation (chain of $n = 4$ agents; 40 natural runs, 30 replays per arm and edge; same receivers, assignments and judge in both replay arms). On Nova Pro—the most resistant model in Table 1—the canonical form is at the floor on *every* edge (0 of 30) while the in-context form is not, so a benchmark using the canonical form reads the model as inert throughout; its self-check fails on two of three edges, which we do not conceal. Claude propagated past the entry edge in no natural run (supplementary Appendix F).

model	edge	in-chain s^{nat}	replay in-context	replay canonical	form effect
Llama	a0→a1	1.000	1.000	1.000	0.000
	a1→a2	0.950	0.967	0.067	+0.900
	a2→a3	0.947	0.733	1.000	−0.267
Nova	a0→a1	0.175	0.067	0.000	+0.067
	a1→a2	0.571	0.333	0.000	+0.333
	a2→a3	0.250	0.500	0.000	+0.500

separate *form* from *regime* we replay both content sources through the *same* harness: a fresh receiver, the same rotating assignment, the same judge. The harness is self-validating, because one arm replays content captured from real natural runs and must reproduce s^{nat} .

Table 4 shows that payload form is a first-order determinant of per-hop survival. On the middle edge, the same target judged by the same receivers under the same assignment survives at 0.967 embedded in a work product and at 0.067 as a bare assertion—a 14× difference produced entirely by packaging. The effect is not uniform across edges: on the last edge the canonical form is *more* effective (1.000 versus 0.733) and the self-check misses (0.733 versus 0.947), so we do not treat that edge as a clean comparison. The frontier model’s block in Table 4 runs against its own headline rate: a benchmark using the canonical form reads the most resistant model as inert throughout.

5.5 Defence: redaction blocks literal attacks, and the bypass depends on headroom

Redaction removes the literal target and drives chain survival to 0.000 on all three models where we measured it. The obfuscation arm (Fig. 2) shows why this is weaker than it appears: on the model with a high no-defence baseline, redaction takes the plain attack from 1.00 to 0.00 while the split attack remains at 1.00 and the spaced attack at 0.88, so it works on the literal channel and is bypassed by ordinary obfuscation. On the resistant frontier model the plain attack succeeds only 0.10 of the time end-to-end (its single-hop cell is 0.17), so redaction’s apparent perfection is the absence of anything to defend (invariant for $\tau_{ASV} \in \{0.7, 0.8, 0.9, 1.0\}$, supplementary Appendix C). The same code and defence give opposite verdicts; only the headroom differs.

Semantic paraphrase is worse than useless in our cells: on the model where the attack succeeds it leaves ASR unchanged (0.450 $\rightarrow$ 0.450) while reducing utility retention from 0.600 to 0.400, so it is *strictly dominated* by doing nothing; on the resistant model it still

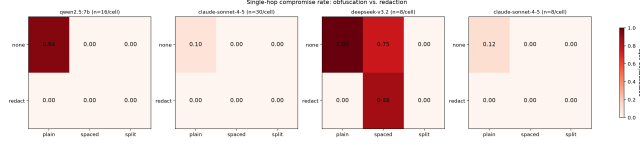


Figure 2: Single-hop compromise rate by defence and attack form, one panel per model ($n = 30$ per cell for Claude 4.5, $n = 8$ for DeepSeek, $n = 16$ for qwen; we do not interpret small differences in the $n = 8$ cells). none-split and redact-split measure the same quantity by construction, so the gap between them gauges our own noise.

costs 0.100 of utility where ASR is already 0.000. This matches prior single-agent findings that paraphrasing is breakable [39].

Both verdicts cover two defences against a static, non-adaptive attacker, and the small panels serve only the within-model contrast. The frontier model’s floor is measured on the bare and plain forms—the least effective ones we test (Table 4)—so we do not claim general resistance.

5.6 Topology and depth

Topology changes not only how much survives but where. On a seven-agent star with the entry at a leaf the centre is compromised at $ASR = 0.725$ with $R_0 = 0.420$ against a regular-topology target $d \cdot \bar{s} = 0.652$ —a structural gap, since the aggregator has no successors—and comparing three seven-agent topologies at nearly equal mean per-hop survival separates *reach* from *reproduction*: the star reaches its aggregation point more often than the chain (0.725 vs. 0.450) while reproducing less (0.420 vs. 0.821), and the tree is the most permissive on both (0.800, 0.831) and the only cell where R_0 approaches $d \cdot \bar{s}$ (within 5%). A topology also decides how much role heterogeneity the network averages over—chain survival spans 0.27–1.00 across receiver roles while all six star edges are worker-to-aggregator and nearly homogeneous—and the star-tree relation itself *reverses with scale* (star above tree on DeepSeek and on a second Llama instance at $n = 10$, tree above star at $n = 7$); the per-topology table is in the supplementary material.

The relative error of the isolated product grows with depth at a model-specific rate, and that rate—not the error at any single depth—is the usable diagnostic. Across models the slopes form a *spectrum* rather than two categories: +11.8 points per hop on qwen2.5:7b (Spearman $\rho = +1.00$, permutation $p = 0.003$, bootstrap interval [+7.8, +14.8]), +6.8 on Llama 3.3 70B ($\rho = +0.77$, $p = 0.014$) before its chain dies at depth 11, and +2.3 on DeepSeek V3.2 at $n = 10$ ($\rho = +0.65$, $p = 0.067$)—a slope whose interval spans zero, so we do not read it as a trend. The ordering follows from what Section 5.2 measures directly: where the isolated rate transports best (DeepSeek matches its measured ASR to 2.2%) composing edge estimates adds little systematic error, whereas an edge under-estimated by 3.75× (Llama’s middle edge) compounds with depth. The slope therefore diagnoses transportability while its *direction* stays model- and topology-dependent. Independence is not what fails here: evaluated on in-chain rates, Eq. (3) reproduces the measured ASR in all three topologies, which is expected rather than informative, since each has effectively one entry–target path and

the recurrence collapses to the chain identity; on a reconvergent DAG, where the two can differ, it underestimates reachability by 2.6× when built from isolated marginals but not from in-chain ones (supplementary Appendix N).

5.7 Recurrence makes the threshold non-vacuous

Section 3.8 shows the threshold criterion vacuous for acyclic graphs and informative only once the interaction graph contains a cycle. We test that second half on a manager-with-reporting-workers pattern: the spectral radius reduces to $\rho(M) = \sqrt{\sum_j a_j c_j}$, yielding *opposite* mean-field predictions borne out by measurement—the Llama and DeepSeek loops are supercritical ($\sum_j a_j c_j = 1.800$ and 1.018) while qwen’s decays (0.578; supplementary Appendix E). We report this as a mean-field prediction on three loops rather than a design rule: the sum counts expected two-step paths, whereas the manager is re-compromised at most once per round, so the exact finite-state prediction is $1 - \prod_j (1 - a_j c_j) < 1$ even above criticality, and the 1.018 loop sits close enough to it that its verdict should be read with the measured dynamics rather than from the point estimate.

5.8 Utility, power and multiplicity

Three facts govern how every null in this paper should be read; their detail, together with the (ASR, ΔU) utility pairs and the estimator-validation battery, is in the supplementary material. At $n = 40$ only deviations above 0.26 are excluded, so no $n = 40$ null is evidence of transport and the degenerate $ASR = \prod_i \hat{s}_i = 0$ cell is uninformative. At $n = 200$ the Llama failure is powered (MDE = 0.11, $\Delta = 0.615$, $p = 0.0002$) and DeepSeek stays a non-rejection at MDE = 0.13 ($\Delta = 0.022$, $p = 0.60$), while a Benjamini–Hochberg correction over the five non-degenerate chain cells leaves only Llama standing. A homogeneity test across the three benign task contexts rejects exchangeability on the weak edge ($\chi^2 = 13.40$, df = 2, $p = 0.0014$ by the Wilson–Hilferty approximation to the χ^2 tail, counts 18/56, 3/56, 13/48), so our rates are context averages rather than i.i.d. draws.

6 Limitations and Threats to Validity

Our coverage is narrow—one benign payload, a static non-adaptive attacker, four to fifteen agents, cells of 30–40 (MDE ≈ 0.26), two defence families—and the utility metric is a contamination proxy rather than a correctness oracle. Protocol sensitivity, the mixed artefact protocol of Table 1, and the scope of the judge are itemised in supplementary Appendix F and supplementary Appendix I.

7 Conclusion

A compromise’s reach is a property of the graph, the roles travelled and the form in which the payload is re-packaged at each hand-off: composition holds in-chain, isolated estimates need not transport, and payload form is a first-order mechanism. On acyclic graphs the epidemic threshold is satisfied identically, so reachability comes from Eq. (3) rather than R_0 . The mechanism serves attackers as well as defenders, so we judge the defensive reading more actionable—*measure the hand-off you will actually run, with an undefended baseline*—and we release no adaptive attacker or tool-calling exploit.

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